

Normalization Confluence in Federated Registry Networks

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Abstract

Normalization confluence [1] guarantees coordination-free convergence within a single registry: under well-founded compensation (WFC) and compensation commutativity (CC), all processors consuming the same events reach the same valid state regardless of application order. We extend this guarantee to *federated* environments where multiple registries, each with independent invariants and compensation, are connected by *registry morphisms* encoding cross-organizational semantic constraints.

Our key construction reduces the federated problem to the single-registry case: a federation of registries connected by morphisms *is* a registry, with morphism consistency conditions as additional invariants and morphism repair as additional compensation. For *tree-shaped* morphism networks (a directed forest: each non-root registry has exactly one incoming morphism), federated convergence requires only validity preservation of the morphisms—all other conditions are *derived* from the network’s acyclicity and the components’ existing convergence properties. The core mechanism is an *authority argument*: in a directed morphism, the source’s unique normal form deterministically determines the target’s shared component, eliminating the nondeterminism that would otherwise require additional axioms.

We prove both structural conditions necessary by counterexample: without acyclicity, contradictory cyclic morphisms cause compensation to cycle; without validity preservation, morphism repair and local compensation oscillate. We then remove the tree restriction altogether: admitting any acyclic network in which a multi-source target carries a *resolution operator*—a source-determined, validity-preserving merge of its incoming constraints—we recover the same convergence guarantee, with single-source authority as the special case. Because the merge encodes domain policy rather than mathematics, the guarantee is conditioned on two decidable properties of the operator, checkable by finite enumeration. For self-containment, we restate the single-registry model and convergence results of [1] alongside a verification calculus that reduces the global CC obligation to local, per-event-pair checks via invariant footprints and decomposable repair.

Keywords: Stream Processing, Normalization Confluence, Semantic Consistency, Rewrite Systems, Newman’s Lemma, Compensation Commutativity, Federated Systems, Registry Morphisms, Distributed Systems

1 Introduction

Coordination-free convergence—the ability of distributed processors to agree on a result without explicit synchronization—is among the central goals of distributed systems design. The literature has identified two structural regimes under which it is achievable:

1. **Operation commutativity** (CRDTs [3]): Operations are designed to commute algebraically. Any application order produces the same state.
2. **Invariant confluence** (I-confluence [6]): Operations individually preserve application invariants under all orderings. No repair is needed because validity is never violated.

Both regimes achieve convergence by *preventing* inconsistency. CRDTs prevent it through algebraic design; I-confluence prevents it through invariant-preserving operation selection. Neither accommodates operations that are non-commutative *and* may violate invariants—yet such operations are pervasive in practice. An order processing pipeline, for example, must handle events (orders, payments, inventory adjustments, shipments) that do not commute and that individually violate business rules (no shipment without payment, no negative inventory).

We formalize the missing piece as a *registry*: any authoritative definition of validity external to the stream processor. Registries already exist in practice under various names—policy engines, constraint services, schema validators, compliance controllers—but their role in convergence has not been characterized formally. Our model abstracts these as a tuple of invariant predicates and a compensation operator.

This paper presents a unified treatment of a third regime, **normalization confluence** [1], and extends it to multi-registry environments. In normalization confluence, operations may be non-commutative and may violate invariants. But a compensation operator normalizes states to validity after each operation, and the key structural condition is that these *compensated results* commute—even though the operations themselves do not. Under this condition, together with well-founded compensation, Newman’s Lemma guarantees that all causality-respecting processing orders reach the same valid state.

The three regimes form a hierarchy of progressively weaker requirements:

1. Operation commutativity $\implies$ compensation commutativity $\implies$ convergence.
2. Invariant confluence $\implies$ no compensation needed $\implies$ convergence.
3. Compensation commutativity alone $\implies$ convergence [1].

Normalization confluence occupies the design space between CRDT commutativity (which requires operations to commute) and I-confluence (which requires operations to preserve invariants). It permits operations that satisfy neither, provided their compensated results agree.

Modern distributed systems, however, span organizational boundaries: a supply chain links manufacturers and suppliers, each with its own registry. We show that normalization confluence extends naturally to these *federated* settings. Multiple registries connected by *morphisms*—structured mappings encoding cross-organizational constraints—form a single federated registry whose convergence reduces to the single-registry case. For tree-shaped morphism networks, this reduction requires surprisingly little beyond the component conditions: only validity preservation of the morphisms. We then lift the tree restriction to any acyclic network by equipping multi-source targets with *resolution operators*—source-determined, validity-preserving merges that generalize the single-source authority function—so convergence holds across arbitrary directed-acyclic constraint graphs.

Relationship to prior work. This paper extends [1], which establishes the single-registry governance rewrite system, the core convergence result ($\text{WFC} + \text{CC} \Rightarrow \text{unique normal forms}$), necessity and complexity results, and a verification calculus for CC based on invariant footprints and decomposable repair. The present paper adds the federated extension (Section 8): registry morphisms with partial authority, the federated construction, the authority argument, the federated convergence theorem with necessity counterexamples, and its generalization to multi-source acyclic networks via resolution operators (Section 8.5). For self-containment, we restate the single-registry model and include all proofs needed to support the federated result; readers familiar with [1] may proceed directly to Section 8.

1.1 Convergence via Normalization

Convergence is conventionally treated as a property of *operations*: operations must be designed to commute, or designed to preserve invariants. Our result reframes convergence as a property of the *normalization rewrite system*: the system that interleaves event application with invariant repair. The operations themselves need no special algebraic structure. What matters is that the rewrite system—encompassing both application and compensation—is terminating and confluent. This shifts the design burden from operation algebra to compensation design.

1.2 Contributions

The single-registry results (items 1–6) are from [1], restated here for self-containment. The federated extension (items 7–10) is new.

1. **Normalization confluence regime** (Section 3): A governance rewrite system on configurations (σ, B) pairing a state with a pending-event buffer, with rules for both event application and compensation. Nondeterminism arises from the choice of which enabled event to apply next.
2. **Convergence theorem** (Section 5): Proof that processors consuming the same events converge to the same valid state, regardless of application order, under well-founded compensation (WFC) and compensation commutativity (CC).
3. **Necessity** (Section 6): Counterexamples demonstrating that without UBC, compensation depth grows without uniform bound (necessary for complexity guarantees), and without CC, different orders reach different valid states (necessary for convergence).
4. **Complexity analysis** (Section 7): Proof that registry governance adds $O(\log n)$ overhead per event under UBC, preserving practical scalability.
5. **Three-regime characterization** (Section 9): Together with CRDTs and invariant confluence, a partition of the coordination-free design space into operation commutativity, invariant confluence, and normalization confluence.
6. **Verification calculus** (Section 10): Sufficient conditions—invariant footprints and decomposable repair—that reduce the global CC obligation to local, per-event-pair checks. Product composition enables modular verification of independent registries.
7. **Federated convergence** (Section 8): Extension to tree-shaped multi-registry networks connected by morphisms. Registry morphisms with partial authority model cross-organizational constraints. Federated WFC and CC are derived from network structure via an authority argument; only morphism validity preservation is needed. Necessity of both acyclicity and validity preservation is shown by counterexample.
8. **Multi-source resolution** (Section 8.5): Removal of the tree restriction. A target with several incoming morphisms carries a *resolution operator* generalizing the authority function; under source determinacy and validity preservation (R1–R2), convergence holds on any

acyclic network (Theorem 8.19), with the tree theorem as the single-source special case. Both conditions are decidable by finite enumeration, so a resolved network is certified exactly as WFC and CC are.

9. **Monotone cycles** (Section 8.6): When shared domains are ordered lattices and repair is monotone, acyclicity is unnecessary—federated convergence holds on *any* network (Theorem 8.24) by the Knaster–Tarski theorem, with order-independence identified as chaotic iteration. CRDTs are the compensation-free special case, and acyclicity is revealed as a concession required only for non-monotone repair.
10. **Compositionality** (Section 8.7): A convex, internally convergent sub-federation collapses to a single *effective registry* (Theorem 8.31); the outer network converges iff the collapsed one does. This gives modular and hierarchical verification—and black-box reuse of verified subsystems—generalizing product composition from independent registries to arbitrary sub-diagrams (limits compose).
11. **Machine-checked base theorem** (Remark 5.4): The single-registry Convergence Theorem this work builds on is mechanized in Coq/Rocq—Newman’s Lemma and the three critical-pair cases—axiom-free (**Print Assumptions** reports “Closed under the global context”), with a concrete registry witnessing that WFC and CC are jointly satisfiable.

2 Related Work

Conflict-Free Replicated Data Types. CRDTs [3] guarantee convergence through algebraic structure: replicas that apply the same set of commutative, associative, idempotent operations converge to the same state. Our setting is strictly more general in one dimension: we do not require operations to commute, only that their *compensated results* commute. This permits operations that individually violate invariants, which CRDTs cannot accommodate. Conversely, CRDTs require no finiteness or boundedness assumptions. Neither approach dominates: CRDTs are preferable when operations can be designed to commute, while registry governance is preferable when business rules constrain valid states in ways that preclude commutativity.

CALM and Coordination Avoidance. The CALM theorem [4, 5] establishes that monotone programs can achieve consistency without coordination. Bailis et al. [6] extend this to identify *invariant confluence*: a set of operations is invariant-confluent if all possible orderings preserve application invariants. When invariant confluence holds, no compensation is needed. Our framework addresses the complement: operations that are *not* invariant-confluent. We prove that compensation restores convergence under bounded depth and commutativity. Together, invariant confluence and our result partition the design space (Section 9.1).

Abstract Rewrite Systems. Newman’s Lemma [2] states that a terminating abstract rewrite system is confluent if and only if it is locally confluent. The lemma has been applied extensively in term rewriting [7] and lambda calculus. Our contribution is its application to a rewrite system with *genuine nondeterminism*: the choice of which event to apply next from a buffer of enabled events. Local confluence is nontrivial because different application orders produce different intermediate states; our compensation commutativity axiom is the condition that closes the resulting critical pairs.

Eventual Consistency. Classical eventual consistency [8] guarantees that replicas converge given sufficient time without new updates. It does not address whether converged states satisfy

application-level invariants—only that replicas agree on the same value. Our work guarantees convergence to *valid* states under continuous event arrival.

Stream Processing Theory. Formal models of stream processing include synchronous dataflow [10], Kahn process networks [11], and the windowed computation models underlying modern engines [12]. These focus on operational semantics—what a stream processor computes—rather than semantic validity of computed states. Our registry layer sits above any of these operational models.

Saga Pattern and Compensating Transactions. The saga pattern [9] decomposes long-lived transactions into sequences of local transactions with compensating actions for rollback. Our compensation operator generalizes this: rather than undoing individual operations, it repairs arbitrary invariant violations in the current state. The convergence guarantee we prove—that different interleaving orders of application and compensation reach the same valid state—has no analogue in the saga literature.

3 Registry-Governed Stream Model

3.1 Registries and Semantic Projection

Definition 3.1 (Registry). A *registry* is a tuple $R = (\Sigma_R, I_R, V_R, \rho_R)$ where:

- Σ_R is a set of *semantic states*,
- $I_R = \{\psi_1, \dots, \psi_k\}$ is a finite set of *invariant predicates* $\psi_i : \Sigma_R \rightarrow \{\top, \perp\}$,
- $V_R : \Sigma_R \rightarrow \{\top, \perp\}$ is the *validation function* defined by $V_R(\sigma) = \bigwedge_{i=1}^k \psi_i(\sigma)$,
- $\rho_R : \Sigma_R \rightarrow \Sigma_R$ is a *deterministic compensation operator* such that if $V_R(\sigma) = \top$, then $\rho_R(\sigma) = \sigma$.

Definition 3.2 (Semantic Projection). A registry is defined over a semantic state space Σ_R together with a *projection* $\pi : \Sigma_{\text{raw}} \rightarrow \Sigma_R$ from unbounded raw states to semantic states. All invariants, validation, and compensation are defined on Σ_R . Two raw states s_1, s_2 are *registry-indistinguishable* if $\pi(s_1) = \pi(s_2)$. The application function $\text{apply} : \mathcal{E} \times \Sigma_R \rightarrow \Sigma_R$ is understood as “apply then project”: given a raw-level application $\text{apply}_{\text{raw}} : \mathcal{E} \times \Sigma_{\text{raw}} \rightarrow \Sigma_{\text{raw}}$, the semantic application is $\text{apply}(e, \sigma) = \pi(\text{apply}_{\text{raw}}(e, s))$ for any s with $\pi(s) = \sigma$. Well-definedness requires that the projected result is independent of the representative s .

Remark 3.3 (Scope of Finiteness). Finiteness is required of the *semantic* state space Σ_R , not the raw data. An order-processing registry tracking status across states *pending*, *paid*, *shipped*, *delivered*, *cancelled* has $|\Sigma_R| = 5$ regardless of how many bytes the underlying order record occupies. More generally, any domain whose logic is captured by k boolean predicates has $|\Sigma_R| \leq 2^k$: the semantic states are the equivalence classes under the invariant vector $(\psi_1(\sigma), \dots, \psi_k(\sigma))$. Business domains are overwhelmingly of this form. The finiteness assumption applies to these semantic equivalence classes, not to raw data representations.

Definition 3.4 (Registry Equivalence). Two states $\sigma_1, \sigma_2 \in \Sigma_R$ are *registry equivalent*, written $\sigma_1 \approx_R \sigma_2$, if they satisfy the same invariants: $\forall \psi \in I_R : \psi(\sigma_1) = \psi(\sigma_2)$.

Lemma 3.5. *Registry equivalence $\approx_R$ is an equivalence relation on Σ_R .*

Proof. Each condition $\psi(\sigma_1) = \psi(\sigma_2)$ is reflexive, symmetric, and transitive. A conjunction of equivalence relations is an equivalence relation. $\square$

3.2 Well-Founded and Uniformly Bounded Compensation

Termination and complexity require two properties of the compensation operator, which we state separately because they play different roles.

Axiom 3.6 (Well-Founded Compensation (WFC)). There exists a *compensation measure* $\Phi : \Sigma_R \rightarrow \mathbb{N}$, defined over semantic states (after projection), such that every compensation step strictly decreases Φ : if $V_R(\sigma) = \perp$, then $\Phi(\rho_R(\sigma)) < \Phi(\sigma)$.

WFC guarantees that compensation terminates from any state: iterated application of ρ_R reaches a valid state in finitely many steps.

Axiom 3.7 (Uniformly Bounded Compensation (UBC)). The compensation measure Φ from Axiom 3.6 is uniformly bounded: there exists $M \in \mathbb{N}$ such that $\Phi(\sigma) \leq M$ for all $\sigma \in \Sigma_R$.

UBC strengthens WFC by bounding compensation depth *uniformly* across all states. UBC does not require finiteness of Σ_R ; bounded compensation height alone suffices. WFC alone suffices for termination and for defining iterated compensation ρ_R^* . UBC is additionally needed for the uniform step bound in the termination lemma and the constant per-event compensation cost in the complexity analysis.

Lemma 3.8 (Finite State Spaces Imply UBC). *If $|\Sigma_R| < \infty$ and WFC holds, then UBC holds with $M = \max_{\sigma \in \Sigma_R} \Phi(\sigma)$.*

Proof. The maximum exists because Σ_R is finite. □

Remark 3.9. WFC is the load-bearing condition for termination; UBC is the load-bearing condition for uniform step bounds and per-event complexity. Finiteness of Σ_R is a convenient sufficient condition for both, but UBC does not require full finiteness of Σ_R ; it requires only bounded compensation height. The counterexample in Section 6 satisfies WFC (compensation terminates for every fixed event set) but violates UBC (compensation depth grows without bound as event sets grow).

3.3 Events and Streams

Definition 3.10 (Event Stream). An *event stream* $\mathcal{S} = \langle e_1, e_2, \dots \rangle$ is a (possibly infinite) sequence of events drawn from event space $\mathcal{E}$. Each event e_i carries a unique identifier $\text{id}(e_i)$, a timestamp $\tau(e_i)$, and a set of causal dependencies $\text{deps}(e_i) \subseteq \{e_1, \dots, e_{i-1}\}$. An *application function* $\text{apply} : \mathcal{E} \times \Sigma_R \rightarrow \Sigma_R$ maps each event and current state to a new state.

Definition 3.11 (Causality-Respecting Order). An ordering π of a finite event set E *respects causality* if whenever $e_i \in \text{deps}(e_j)$, event e_i precedes e_j under π . Multiple causality-respecting orders may exist for the same event set.

3.4 Governance Rewrite System

We now define the rewrite system that is the core technical device of this paper. Unlike the approach of fixing a canonical event order—which would make agreement trivial by determinism—our rewrite system admits genuine nondeterminism in the choice of which event to apply next.

Assumption 3.12 (Causal Closure). The received event set E is causally closed: for every $e \in E$, $\text{deps}(e) \subseteq E$.

Under causal closure, every dependency of every event in E is either still pending in B or has already been applied (removed from B). This makes the enabled predicate below well-defined.

Definition 3.13 (Configuration). A *configuration* is a pair (σ, B) where $\sigma \in \Sigma_R$ is a semantic state and $B \subseteq E$ is a set of events that have been received but not yet applied. An event $e \in B$ is *enabled* if $\text{deps}(e) \cap B = \emptyset$ (all causal dependencies of e have already been applied).

Definition 3.14 (Governance Rewrite System). Fix a finite set of received events E and initial state σ_0 . The *governance rewrite system* $\mathcal{G} = (\mathcal{C}, \rightarrow)$ operates on configurations (σ, B) with $\sigma \in \Sigma_R$ and $B \subseteq E$. The rewrite relation $\rightarrow$ is the union of:

1. **Apply step:** $(\sigma, B) \rightarrow (\text{apply}(e, \sigma), B \setminus \{e\})$ whenever $e \in B$ is enabled.
2. **Compensation step:** $(\sigma, B) \rightarrow (\rho_R(\sigma), B)$ whenever $V_R(\sigma) = \perp$.

A configuration is in *normal form* if $B = \emptyset$ and $V_R(\sigma) = \top$.

The nondeterminism is genuine: when multiple events are enabled and the state is invalid, a configuration may step via different apply rules (choosing different events) or via the compensation rule. Confluence of this system means that the choice does not affect the final result.

3.5 Compensation Commutativity

Local confluence of $\mathcal{G}$ is nontrivial. Two critical pair types arise: (1) two apply steps choosing different events, and (2) an apply step versus a compensation step. The following axiom ensures both types close.

Definition 3.15 (Iterated Compensation). Under WFC (Axiom 3.6), define $\rho_R^*(\sigma)$ to be the unique state obtained by iterating ρ_R from σ until reaching validity: $\rho_R^*(\sigma) = \rho_R^m(\sigma)$ for the least m such that $V_R(\rho_R^m(\sigma)) = \top$. Existence and finiteness of m follow from WFC; uniqueness follows from determinism of ρ_R .

Axiom 3.16 (Compensation Commutativity (CC)). The registry R satisfies:

(CC1) Order independence. For any $\sigma \in \Sigma_R$ and events e_1, e_2 that are causally independent and simultaneously enabled in some configuration (σ, B) :

$$\rho_R^*(\text{apply}(e_2, \rho_R^*(\text{apply}(e_1, \sigma)))) = \rho_R^*(\text{apply}(e_1, \rho_R^*(\text{apply}(e_2, \sigma)))).$$

(CC2) Compensation absorption. For any $\sigma \in \Sigma_R$ with $V_R(\sigma) = \perp$ and any enabled event e :

$$\rho_R^*(\text{apply}(e, \sigma)) = \rho_R^*(\text{apply}(e, \rho_R(\sigma))).$$

Remark 3.17 (CRDT Commutativity). CRDT commutativity requires $\text{apply}(e_1, \text{apply}(e_2, \sigma)) = \text{apply}(e_2, \text{apply}(e_1, \sigma))$ for all σ . This implies CC1 (take $\rho_R^* = \text{id}$ when all states are valid) and trivially satisfies CC2 (when all states are valid, compensation never fires). Compensation commutativity is strictly weaker: it permits operations that individually violate invariants, requiring only that the *repaired* results commute. The gap between operation commutativity and compensation commutativity is precisely the space our result occupies.

Remark 3.18 (Compensation Absorption). Condition CC2 states that compensating an invalid state before applying an event does not change the final compensated result. Intuitively, the repair that compensation performs “would have been performed anyway” during the post-application compensation phase. CC2 holds whenever compensation repairs invariant violations in a way that is determined by the violation pattern rather than the path taken to reach the violation. By induction on the number of compensation steps, CC2 generalizes: $\rho_R^*(\text{apply}(e, \sigma)) = \rho_R^*(\text{apply}(e, \rho_R^*(\sigma)))$. In rewrite-theoretic terms, CC2 ensures that the normalizer is stable under interleaving with event application: normalizing “early” (before an event) versus “late” (after) does not change the final normal form.

3.6 Stream Processors

Definition 3.19 (Stream Processor). A *stream processor* P for stream $\mathcal{S}$ under registry R is a process that, at each time t , has received some subset of events $E_P(t) \subseteq \mathcal{S}$ and maintains state $\Psi_P(t) \in \Sigma_R$. We require:

1. **Eventual delivery:** For every event $e_i \in \mathcal{S}$, there exists a time t_i such that $e_i \in E_P(t)$ for all $t \geq t_i$.
2. **Causality-respecting application:** The processor applies events in some order that respects causality (Definition 3.11), buffering events whose causal dependencies have not yet been received. The processor is free to choose any causality-respecting order.
3. **Incremental compensation:** After applying each event, the processor applies ρ_R until validity is restored.

The processor's computation on event set $E_P(t)$ is a reduction sequence in the governance rewrite system $\mathcal{G}$ starting from $(\sigma_0, E_P(t))$.

Different processors may apply the same events in different causality-respecting orders. The convergence theorem guarantees they nonetheless arrive at the same valid state.

4 Worked Example: Order Fulfillment

We instantiate the framework on an order fulfillment pipeline, verify the structural conditions, and trace two processor orderings to the same valid state.

4.1 Registry Definition

An order has a *stage* in $\{\text{pending}, \text{approved}, \text{held}\}$ and a *balance* in $\{0, 1, 2, 3\}$. The semantic state space is $\Sigma_R = \{\text{pending}, \text{approved}, \text{held}\} \times \{0, 1, 2, 3\}$, with $|\Sigma_R| = 12$ and initial state $\sigma_0 = (\text{pending}, 0)$.

A single invariant governs fulfillment:

$$\psi(s, b) = \top \iff (s = \text{approved} \implies b \geq 2).$$

Approval requires a balance of at least 2. The only invalid states are $(\text{approved}, 0)$ and $(\text{approved}, 1)$.

Compensation. When an order is approved with insufficient balance, it is placed on hold:

$$\rho(\text{approved}, b) = (\text{held}, b) \quad \text{for } b < 2.$$

All other states are valid, and ρ acts as the identity. The *held* stage preserves the intent to approve: the system records that approval was requested but cannot yet proceed. This is a deliberate design choice—a naive alternative that simply cancels the approval (reverting to *pending*) would violate CC, as we discuss below.

Measures. $\Phi(s, b) = 1$ if $s = \text{approved}$ and $b < 2$, otherwise 0. WFC holds (ρ strictly decreases Φ on invalid states), and UBC holds with $M = 1$.

4.2 Events

Two causally independent events:

- e_{credit} : Adds 1 to the balance (capped at 3). If the order is on hold and the new balance reaches 2, the order is automatically approved:

$$\text{apply}(e_{\text{credit}}, (s, b)) = \begin{cases} (\text{approved}, \min(b+1, 3)) & \text{if } s = \text{held} \text{ and } \min(b+1, 3) \geq 2, \\ (s, \min(b+1, 3)) & \text{otherwise.} \end{cases}$$

- e_{approve} : Sets the stage to *approved*: $\text{apply}(e_{\text{approve}}, (s, b)) = (\text{approved}, b)$.

4.3 Verifying CC

CC1 (Order Independence). We verify CC1 for $(e_{\text{approve}}, e_{\text{credit}})$ at two representative starting states.

From $(\text{pending}, 0)$:

Path 1 (e_{approve} first):

$$(\text{pending}, 0) \xrightarrow{e_{\text{approve}}} (\text{approved}, 0) \xrightarrow{\rho^*} (\text{held}, 0) \xrightarrow{e_{\text{credit}}} (\text{held}, 1). \quad \text{Result: } (\text{held}, 1).$$

Path 2 (e_{credit} first):

$$(\text{pending}, 0) \xrightarrow{e_{\text{credit}}} (\text{pending}, 1) \xrightarrow{e_{\text{approve}}} (\text{approved}, 1) \xrightarrow{\rho^*} (\text{held}, 1). \quad \text{Result: } (\text{held}, 1). \quad \checkmark$$

In Path 1, approval fires first, violates the invariant (balance $0 < 2$), and compensation places the order on hold. The subsequent credit increments the balance but does not yet reach the threshold. In Path 2, credit arrives first (no violation), then approval violates the invariant and compensation again produces the held state. Both paths reach $(\text{held}, 1)$.

From $(\text{pending}, 1)$:

$$\text{Path 1: } (\text{pending}, 1) \xrightarrow{e_{\text{approve}}} (\text{approved}, 1) \xrightarrow{\rho^*} (\text{held}, 1) \xrightarrow{e_{\text{credit}}} (\text{approved}, 2). \quad \text{Result: } (\text{approved}, 2).$$

$$\text{Path 2: } (\text{pending}, 1) \xrightarrow{e_{\text{credit}}} (\text{pending}, 2) \xrightarrow{e_{\text{approve}}} (\text{approved}, 2). \quad \text{Result: } (\text{approved}, 2). \quad \checkmark$$

In Path 1, approval again violates the invariant; compensation holds; the subsequent credit reaches balance 2, triggering auto-approval. In Path 2, the credit arrives first, and approval at balance 2 is valid. Both reach $(\text{approved}, 2)$: the order is approved with sufficient funds.

Since $|\Sigma_R| = 12$, CC1 can be verified exhaustively for all valid starting states. All cases follow the same pattern: compensation preserves the approval intent in the held state, and subsequent credits eventually satisfy the balance threshold.

CC2 (Compensation Absorption). From the invalid state $(\text{approved}, 0)$ with event e_{credit} :

$$\text{LHS: } \rho^*(\text{apply}(e_{\text{credit}}, (\text{approved}, 0))) = \rho^*((\text{approved}, 1)) = (\text{held}, 1).$$

$$\text{RHS: } \rho^*(\text{apply}(e_{\text{credit}}, \rho(\text{approved}, 0))) = \rho^*(\text{apply}(e_{\text{credit}}, (\text{held}, 0))) = \rho^*((\text{held}, 1)) = (\text{held}, 1). \quad \checkmark$$

Compensating before the credit (putting the order on hold first) produces the same result as applying the credit to the invalid state and then compensating. CC2 holds because the held state and the approved-but-invalid state respond identically to subsequent credits after compensation.

4.4 Two Processors, Three Events

Consider the stream $\mathcal{S} = \langle e_{\text{approve}}, e_{\text{credit}}, e_{\text{credit}} \rangle$ (approval followed by two credits). Two processors P_1, P_2 receive the events in different orders:

Processor P_1 (approve, credit, credit):

$$(pending, 0) \xrightarrow{e_{\text{approve}}} (approved, 0) \xrightarrow{\rho^*} (held, 0) \xrightarrow{e_{\text{credit}}} (held, 1) \xrightarrow{e_{\text{credit}}} (approved, 2).$$

Processor P_2 (credit, credit, approve):

$$(pending, 0) \xrightarrow{e_{\text{credit}}} (pending, 1) \xrightarrow{e_{\text{credit}}} (pending, 2) \xrightarrow{e_{\text{approve}}} (approved, 2).$$

Both processors arrive at $(approved, 2)$. P_1 encountered an invariant violation (approval without sufficient balance) and passed through the held state; P_2 never violated the invariant. Compensation commutativity guarantees agreement regardless of the path.

4.5 Why Compensation Design Matters

The held stage is essential. A naive compensation that simply reverts approval to *pending*—i.e., $\rho(approved, b) = (pending, b)$ —would violate CC1. From $(pending, 0)$, approval-then-credit would produce $(pending, 1)$ (intent to approve lost), while credit-then-approval would produce $(held, 1)$ or a different compensated state depending on the design. The held stage preserves the approval intent so that subsequent credits can trigger auto-approval, ensuring that different orderings converge.

This illustrates the practical content of the CC conditions: they constrain *compensation design*, not operation design. The sufficient condition used here is *state-determined compensation*: ρ^* depends only on which invariants are violated (an approved order with low balance is always placed on hold), making the post-compensation state independent of how the violation was reached.

5 Convergence Theorem

5.1 Termination

Lemma 5.1 (Termination). *Under WFC (Axiom 3.6), the governance rewrite system $\mathcal{G}$ is terminating: every reduction sequence from (σ_0, E) is finite. Under UBC (Axiom 3.7), the total number of steps is bounded by $|E| + (|E| + 1) \cdot M$.*

Proof. Define the measure $\mu(\sigma, B) = (|B|, \Phi(\sigma))$ taking values in $\mathbb{N} \times \mathbb{N}$ under the lexicographic order, which is well-founded.

An apply step sends (σ, B) to $(\text{apply}(e, \sigma), B \setminus \{e\})$. The first component decreases: $|B \setminus \{e\}| = |B| - 1$. Therefore μ strictly decreases regardless of the second component.

A compensation step sends (σ, B) to $(\rho_R(\sigma), B)$. The first component is unchanged; the second strictly decreases by WFC. Therefore μ strictly decreases.

Since μ maps into a well-ordered set and strictly decreases at every step, every reduction sequence is finite. Under UBC, $\Phi(\sigma) \leq M$ for all σ , so at most M compensation steps can occur between apply steps (including before the first apply), giving the uniform bound $|E| + (|E| + 1) \cdot M$. $\square$

5.2 Local Confluence

Lemma 5.2 (Local Confluence). *Under CC (Axiom 3.16), the governance rewrite system $\mathcal{G}$ is locally confluent: if $(\sigma, B) \rightarrow X_1$ and $(\sigma, B) \rightarrow X_2$, then there exists a configuration Z with $X_1 \rightarrow^* Z$ and $X_2 \rightarrow^* Z$.*

Proof. Three cases arise from the two rewrite rules.

Case 1: Two apply steps. $(\sigma, B) \rightarrow (\text{apply}(e_1, \sigma), B \setminus \{e_1\}) = X_1$ and $(\sigma, B) \rightarrow (\text{apply}(e_2, \sigma), B \setminus \{e_2\}) = X_2$, where $e_1 \neq e_2$ are both enabled.

Since e_1 and e_2 are both enabled, their dependencies are disjoint from B . By Assumption 3.12, $e_1, e_2 \in E$, and since $e_1 \in B$ we have $e_1 \notin \text{deps}(e_2)$ (otherwise $\text{deps}(e_2) \cap B \neq \emptyset$, contradicting enabledness of e_2), and symmetrically $e_2 \notin \text{deps}(e_1)$. Therefore e_1 and e_2 are causally independent.

From X_1 : compensate to validity, obtaining $(\rho_R^*(\text{apply}(e_1, \sigma)), B \setminus \{e_1\})$; then apply e_2 (still enabled since its dependencies remain outside $B \setminus \{e_1\}$) and compensate, reaching

$$Z_1 = (\rho_R^*(\text{apply}(e_2, \rho_R^*(\text{apply}(e_1, \sigma)))), B \setminus \{e_1, e_2\}).$$

From X_2 : symmetrically, reach

$$Z_2 = (\rho_R^*(\text{apply}(e_1, \rho_R^*(\text{apply}(e_2, \sigma)))), B \setminus \{e_1, e_2\}).$$

By CC1, the state components of Z_1 and Z_2 are equal. Take $Z = Z_1 = Z_2$.

Case 2: Apply vs. compensation. $V_R(\sigma) = \perp$ and e is enabled in B . $(\sigma, B) \rightarrow (\text{apply}(e, \sigma), B \setminus \{e\}) = X_1$ and $(\sigma, B) \rightarrow (\rho_R(\sigma), B) = X_2$.

From X_1 : compensate to validity, reaching $(\rho_R^*(\text{apply}(e, \sigma)), B \setminus \{e\})$.

From $X_2 = (\rho_R(\sigma), B)$: apply e (still enabled, since compensation does not change B) to get $(\text{apply}(e, \rho_R(\sigma)), B \setminus \{e\})$; then compensate to validity, reaching $(\rho_R^*(\text{apply}(e, \rho_R(\sigma))), B \setminus \{e\})$.

By CC2, $\rho_R^*(\text{apply}(e, \sigma)) = \rho_R^*(\text{apply}(e, \rho_R(\sigma)))$. The buffer components are equal. Take $Z = (\rho_R^*(\text{apply}(e, \sigma)), B \setminus \{e\})$.

Case 3: Two compensation steps. Both rewrites apply ρ_R to the same state, producing the same successor $(\rho_R(\sigma), B)$. Take $Z = (\rho_R(\sigma), B)$. $\square$

5.3 Main Result

The following results are from [1]; proofs are included for self-containment.

Corollary 5.3 (Unique Normal Forms). *Under WFC and CC, every configuration (σ_0, E) has a unique normal form.*

Proof. By Lemma 5.1, $\mathcal{G}$ is terminating. By Lemma 5.2, $\mathcal{G}$ is locally confluent. By Newman’s Lemma [2], $\mathcal{G}$ is globally confluent. A terminating, confluent rewrite system has unique normal forms. $\square$

Remark 5.4 (Machine-checked). The single-registry base results above ([1]) are mechanized in Coq/Rocq (checked with Rocq 9.3): Newman’s Lemma is proved from scratch, and the Governance Rewrite System’s termination (via the lexicographic measure $(|B|, \Phi(\sigma))$) and local confluence (the three critical-pair cases, closed by CC1, CC2, and determinism of ρ_R) yield confluence and unique normal forms. For both the abstract theorem and a concrete instantiation, **Print Assumptions** reports “Closed under the global context”: no axioms, no admitted lemmas. A concrete state-determined-compensation registry discharges WFC and CC with genuine nondeterminism, so the hypotheses are jointly satisfiable and the theorem is not vacuous. The soundness of the reference implementation’s certification strategy is mechanized as well—footprint disjointness implies order-independent commutation (the compositional CC path), and a repair strictly decreasing a potential is strongly normalizing (WFC)—each axiom-free. The monotone-cycles result (Theorem 8.24) is mechanized in its constructive finite-lattice core: Kleene iteration of a monotone operator from $\perp$ is an ascending chain bounded by every fixed point, and once it stabilizes the stable value is the (unique) least fixed point $\bar{s}^*$, again axiom-free with a concrete witness. The chaotic-iteration convergence—that every fair *asynchronous* component-wise schedule from $\perp$, not only the synchronous iteration, reaches $\bar{s}^*$ —is mechanized as well: reframing a chaotic schedule as a productive-update rewrite relation, below $\bar{s}^*$ a monotone update strictly increases a rank, so the relation is strongly normalizing and its unique normal form is $\bar{s}^*$; hence any productive schedule reaches the same fixed point (order-independence), again axiom-free with a concrete witness. The single-registry development is furthermore paired with two runnable checkers extracted from it, each proven axiom-free: one certifies that a compiled machine’s emitted step tables commute and stay in range, and one recomputes convergence directly from the machine’s rules by evaluating their expression trees (trusting neither the implementation’s enumeration nor its normalization). Extracted from the proof rather than reimplemented by hand, they act as independent differential oracles for the reference implementation: a defect in its hand-written verification cannot make a non-convergent machine pass either checker. Scripts are in the `coq/` directory of the artifact.

Theorem 5.5 (Stream Convergence). *Let R be a registry satisfying WFC (Axiom 3.6) and CC (Axiom 3.16), and let P_1, P_2 be stream processors (Definition 3.19) consuming the same event stream $\mathcal{S}$. Then:*

- (a) **Validity:** *For each processor, after it completes the incremental compensation phase for any applied event, its state satisfies $V_R(\Psi_{P_j}(t)) = \top$.*
- (b) **Order independence:** *The state of each processor depends only on its received event set, not on the order in which events were applied.*
- (c) **Agreement:** *Whenever the processors have received the same events, they are in the same state: if $E_{P_1}(t) = E_{P_2}(t)$, then $\Psi_{P_1}(t) = \Psi_{P_2}(t)$.*

Combined with eventual delivery, disagreements are transient: any event causing processors to differ will eventually be received by both, restoring agreement.

Proof. We prove each part.

Part (a): Validity. By Lemma 5.1, compensation terminates. By Definition 3.19, the processor compensates after each event application until validity is restored. Therefore after completing the compensation phase for any applied event, $V_R(\Psi_{P_j}(t)) = \top$.

Part (b): Order independence. Each processor’s computation on event set $E = E_{P_j}(t)$ is a reduction sequence in the governance rewrite system from (σ_0, E) to a normal form $(\sigma^*, \emptyset)$. Different causality-respecting orders correspond to different reduction sequences. By Corollary 5.3, all reduction sequences reach the same normal form. Therefore the final state σ^* depends only on E , not on the order.

Part (c): Agreement. If $E_{P_1}(t) = E_{P_2}(t) = E$, then both processors’ computations are reduction sequences from (σ_0, E) . By Corollary 5.3, both reach the same normal form $(\sigma^*, \emptyset)$. Therefore $\Psi_{P_1}(t) = \Psi_{P_2}(t) = \sigma^*$. $\square$

Remark 5.6 (Semantic Projection). The theorem implies that the mapping $F(E) = \text{nf}(\sigma_0, E)$ —the unique normal form of the governance rewrite system starting from (σ_0, E) —is a well-defined function of the event set E , independent of causal linearization. This is the Church–Rosser property for our operational semantics.

Corollary 5.7 (Convergence Under Quiescent Streams). *If the stream $\mathcal{S}$ is finite, then there exists a valid state σ^* such that both processors converge to σ^* : there exists T with $\Psi_{P_1}(t) = \Psi_{P_2}(t) = \sigma^*$ for all $t \geq T$.*

Proof. By eventual delivery, there exists T_N such that $E_{P_1}(t) = E_{P_2}(t) = \mathcal{S}$ for all $t \geq T_N$. By Theorem 5.5(c), both processors are in the same state. Since no new events arrive, this state is fixed from T_N onward. $\square$

Remark 5.8 (Infinite Streams). For infinite streams, the received event sets grow monotonically toward $\mathcal{S}$ but may differ at any given time. When event sets coincide, Theorem 5.5(c) guarantees agreement. Eventual delivery ensures that any event causing disagreement is transient. The agreed-upon state may continue to change as new events arrive—the guarantee is perpetual agreement, not convergence to a fixed point. Corollary 5.7 recovers fixed-point convergence when the stream is finite.

6 Necessity of the Structural Conditions

We show that both UBC and CC are necessary for their respective guarantees: UBC for uniform complexity, CC for agreement.

6.1 Necessity of UBC

UBC is not merely convenient; it is necessary for a uniform bound on compensation depth, and thus for the constant per-event compensation overhead used in the complexity analysis (Section 7). Without it, compensation depth can grow without limit as event sets grow.

Theorem 6.1 (Unbounded Compensation Depth Without UBC). *For any $M \in \mathbb{N}$, there exists a registry R_∞ and a finite event set E such that some reduction sequence from (σ_0, E) in the governance rewrite system contains more than M compensation steps. Consequently, no uniform bound on compensation depth exists, and UBC fails.*

Proof. Define R_∞ with $\Sigma_{R_\infty} = \mathbb{Z}$, the single invariant $\psi(\sigma) = \top \iff \sigma = 0$, initial state $\sigma_0 = 0$, and compensation $\rho(\sigma) = \sigma + 1$ for $\sigma < 0$ and $\rho(\sigma) = \sigma - 1$ for $\sigma > 0$ (so compensation moves toward 0).

For each $n \in \mathbb{N}$, consider the single-event set $E_n = \{e_n\}$ where $\text{apply}(e_n, 0) = -n$. Under the processor model (Definition 3.19), the processor applies e_n to reach state $-n$, then compensates incrementally: $-n \rightarrow -(n-1) \rightarrow \dots \rightarrow -1 \rightarrow 0$. This requires exactly n compensation steps.

For any proposed bound M , choose $n = M + 1$. The compensation episode after this single event contains $n > M$ steps. Therefore no finite M uniformly bounds compensation depth, and UBC fails.

The measure $\Phi(\sigma) = |\sigma|$ strictly decreases under ρ , so WFC holds and compensation terminates for each fixed E_n . The pathology is not non-termination but *unbounded* termination: the compensation depth required to restore validity grows without limit. $\square$

Remark 6.2 (What Fails). The counterexample satisfies WFC (compensation terminates for every fixed E_n) but violates UBC (compensation depth is n , unbounded). CC also holds: there is only one event per set, so order independence is vacuous, and compensation absorption holds because ρ moves deterministically toward 0. The pathology is purely one of unbounded depth, which invalidates the uniform step bound and the constant per-event compensation cost in Theorem 7.1. Finite semantic state spaces prevent this by bounding Φ (Lemma 3.8).

6.2 Necessity of CC

CC is necessary for agreement: without it, processors applying the same events in different causality-respecting orders can reach different valid states.

Proposition 6.3 (Disagreement Without CC). *There exists a registry satisfying WFC and UBC, and a finite event set E with two causality-respecting orderings, such that the governance rewrite system reaches different valid normal forms under the two orderings.*

Proof. Let $\Sigma_R = \{A, B, C, D\}$ with valid states $\{A, B\}$ and compensation $\rho_R(C) = A$, $\rho_R(D) = B$. UBC holds with $M = 1$.

Define two events e_1, e_2 with $\text{deps}(e_1) = \text{deps}(e_2) = \emptyset$, so both are enabled from the initial configuration $(A, \{e_1, e_2\})$ and are causally independent under Definition 3.11. Their application is:

	A	B	C	D
e_1	C	C	C	D
e_2	D	C	C	D

Evaluate CC1 at $\sigma = A$. Under the processor model (Definition 3.19), each processor normalizes after each application, so the comparison is between $\rho_R^*(A_{e_2}(\rho_R^*(A_{e_1}(A))))$ and the symmetric order.

Path 1 (e_1 then e_2): $\text{apply}(e_1, A) = C$, $\rho_R^*(C) = A$, $\text{apply}(e_2, A) = D$, $\rho_R^*(D) = B$. Normal form: B .

Path 2 (e_2 then e_1): $\text{apply}(e_2, A) = D$, $\rho_R^*(D) = B$, $\text{apply}(e_1, B) = C$, $\rho_R^*(C) = A$. Normal form: A .

Since $A \neq B$, CC1 is violated. Two processors consuming $\{e_1, e_2\}$ in different causality-respecting orders reach different valid states despite WFC and UBC holding. $\square$

Remark 6.4. The key feature is that ρ_R maps distinct invalid states to *distinct* valid states ($C \mapsto A$, $D \mapsto B$), creating path-dependent repair. Contrast this with the order fulfillment example (Section 4), where compensation preserves approval intent via the held state, ensuring path-independent repair. CC captures precisely the boundary between these cases.

Remark 6.5. The UBC counterexample (Theorem 6.1) satisfies CC but violates UBC; the CC counterexample (Proposition 6.3) satisfies WFC and UBC but violates CC. Together, they show that each condition is independently necessary for its respective guarantee.

7 Complexity Analysis

Theorem 7.1 (Convergence Complexity). *For a stream prefix of length n under a registry satisfying UBC with bound M and $k = |I_R|$ invariants, the model-level cost of registry-governed stream processing is $O(n)$ for event application, validation, and compensation combined. If causality maintenance is implemented with a priority queue, the total cost is $O(n \log n)$, where k and M are treated as constants of system design.*

Proof. We analyze the per-event cost, separating model-level costs (which follow from the formalism) from scheduling costs (which depend on implementation).

Model-level per-event costs. For each event e_i ($1 \leq i \leq n$):

- *Event application:* $O(1)$ (state update).
- *Invariant validation:* $O(k) = O(1)$ (evaluate each ψ_j).
- *Compensation:* At most M steps, each $O(1)$. Since M is bounded by UBC, this is $O(1)$.

Registry governance therefore adds $O(1)$ overhead per event beyond what the base stream processor already performs.

Scheduling overhead. Maintaining a set of enabled events and resolving causal dependencies requires implementation-dependent data structures. With a priority queue keyed by timestamp and identifier, each event insertion and dependency-resolution check costs $O(\log i)$, yielding:

Total cost.

$$T(n) = \sum_{i=1}^n (\underbrace{O(1)}_{\text{model}} + \underbrace{O(\log i)}_{\text{scheduling}}) = O(n) + O\left(\sum_{i=1}^n \log i\right) = O(n \log n).$$

The dominant term is causality maintenance. Validation and compensation contribute no asymptotic overhead. Systems with native causal ordering (e.g., single-partition Kafka topics) eliminate the $O(\log i)$ term, reducing the bound to $O(n)$. $\square$

8 Federated Convergence

The convergence theorem (Theorem 5.5) applies within a single registry. The product composition theorem (Theorem 10.13) extends it to independent registries with disjoint invariants. But real multi-registry systems have *dependencies* across organizational boundaries: a manufacturer’s specifications constrain supplier capabilities; a regulator’s rules constrain bank behavior. We now extend convergence to registries connected by morphisms encoding such constraints.

The central observation is that a federation of registries connected by morphisms *is* a registry in the sense of Section 3: the federated state space is the product of component state spaces, morphism consistency conditions are additional invariants, and morphism repair is additional compensation. Federated convergence then follows from Theorem 5.5 if the federated registry satisfies WFC and CC. We prove both are derived from the network’s structure rather than assumed.

8.1 Registry Morphisms

Definition 8.1 (Registry Morphism). Given registries R_A and R_B , a *registry morphism* $\phi : R_A \rightarrow R_B$ consists of a *decomposition* of B 's state space—an isomorphism $\langle \pi_B^{\text{sh}}, \pi_B^{\text{loc}} \rangle : \Sigma_B \xrightarrow{\sim} S_B \times L_B$ with reconstruction $\text{merge}_B : S_B \times L_B \rightarrow \Sigma_B$ satisfying $\text{merge}_B(\pi_B^{\text{sh}}(\sigma), \pi_B^{\text{loc}}(\sigma)) = \sigma$ —and a function $\phi : \Sigma_A \rightarrow S_B$ mapping source states to *shared* target states. Write $\sigma_B[\text{sh} \mapsto s] := \text{merge}_B(s, \pi_B^{\text{loc}}(\sigma_B))$.

The morphism satisfies:

(M1) **Validity preservation under overwrite:** If $V_A(\sigma_A) = \top$ and $V_B(\sigma_B) = \top$, then $V_B(\sigma_B[\text{sh} \mapsto \phi(\sigma_A)]) = \top$.

When $\pi_B^{\text{sh}} = \text{id}$ (L_B trivial), the morphism constrains B completely (*total morphism*).

Remark 8.2. M1 is not standard function-level validity preservation; it is validity preservation under shared-component overwrite. A sufficient condition: if V_B factors as $V_B(\sigma) \iff V_B^{\text{sh}}(\pi_B^{\text{sh}}(\sigma)) \wedge V_B^{\text{loc}}(\pi_B^{\text{loc}}(\sigma))$ (shared and local validity independent), then M1 reduces to $V_A(\sigma_A) = \top \Rightarrow V_B^{\text{sh}}(\phi(\sigma_A)) = \top$. The shared/local decomposition models partial authority: a manufacturer constrains product specifications (shared) but not supplier pricing (local).

M1 is a nonvacuous constraint: its antecedent requires both a valid source and a valid target state, and WFC (Axiom 3.6) guarantees every component registry has a nonempty valid set, since iterated compensation reaches a valid state from any state. A registry whose invariants admit no valid state is therefore excluded at the component level, not silently satisfying M1 here.

Definition 8.3 (Tree-Shaped Morphism Network). A *morphism network* $\mathcal{N} = (\{R_i\}_{i \in \mathcal{I}}, \{\phi_{ij}\}_{(i,j) \in E})$ is *tree-shaped* if $(\mathcal{I}, E)$ is a directed forest: each connected component is a tree rooted at a *source* registry (no incoming morphisms), and each non-source has exactly one incoming edge.

8.2 Federated Construction

Definition 8.4 (Federated Registry). Given a tree-shaped morphism network $\mathcal{N}$, the federated registry R_{Fed} has state space $\Sigma_{\text{Fed}} = \prod_i \Sigma_{R_i}$ with projections $\pi_i(\bar{\sigma}) = \sigma_i$. Invariants comprise: (i) *local invariants*—each $\psi_j \in I_{R_i}$ lifted to the product as $\hat{\psi}_{i,j}(\bar{\sigma}) = \psi_j(\sigma_i)$ —and (ii) *morphism invariants*—for each $(i, j) \in E$, $\mu_{ij}(\bar{\sigma}) = \top \iff \phi_{ij}(\sigma_i) = \pi_j^{\text{sh}}(\sigma_j)$.

Definition 8.5 (Federated Normalization Operator). Let $i_1, \dots, i_n$ be a topological ordering of $\mathcal{I}$. The operator $\rho_{\text{Fed}} : \Sigma_{\text{Fed}} \rightarrow \Sigma_{\text{Fed}}$ acts in two phases:

Phase 1 (Local normalization): For each i , replace σ_i with $\rho_{R_i}^*(\sigma_i)$.

Phase 2 (Morphism repair): Process edges in topological order. For $(i, j) \in E$, if $\phi_{ij}(\sigma_i) \neq \pi_j^{\text{sh}}(\sigma_j)$, set $\sigma_j \mapsto \sigma_j[\text{sh} \mapsto \phi_{ij}(\sigma_i)]$.

Local normalization precedes morphism repair because M1 guarantees that overwriting the shared component of a locally valid target with the morphism image of a locally valid source preserves target validity. Federated events target a specific registry R_m and modify only component m ; cross-registry effects arise through morphism violation and repair. The federated governance rewrite system $\mathcal{G}_{\text{Fed}}$ has apply steps and compensation steps as in the single-registry case (Definition 3.14).

8.3 The Authority Argument

In a directed morphism $\phi : R_A \rightarrow R_B$, registry A is *authoritative* over B 's shared component: the federated normal form of B 's shared state is completely determined by A 's normal form. The federated normal form decomposes into three kinds of state components, each with a separate convergence argument.

Lemma 8.6 (Authority Determination). *In a federally valid state $\bar{\sigma}$:*

- (a) *Each source registry's state equals its single-registry normal form (events targeting other registries and morphism repair do not modify source components).*
- (b) *The shared component of non-source j with incoming edge from i satisfies $\pi_j^{\text{sh}}(\sigma_j) = \phi_{ij}(\sigma_i)$ (determined by source normal forms propagated through morphisms).*
- (c) *The local component of non-source j is unconstrained by morphisms and converges via component CC for R_j .*

Lemma 8.7 (Single-Round Termination). *A single application of ρ_{Fed} produces a federally valid state. Phase 1 achieves local validity in $\leq \sum_i M_i$ steps; Phase 2 processes each edge once (unique target: each registry's shared component is set once and not revisited). Critically, morphism repair cannot reintroduce local invalidity: M1 guarantees that overwriting the shared component of a locally valid target with the morphism image of a locally valid source preserves target validity. Since Phase 1 establishes local validity before Phase 2 begins, and Phase 2 only modifies shared components, local validity is an invariant of the entire procedure.*

Theorem 8.8 (Federated CC). *Let $\mathcal{N}$ be a tree-shaped morphism network where each component satisfies WFC and CC. Then the federated system satisfies CC.*

Proof. We verify CC1 and CC2 by decomposing the federated normal form into three component types per Lemma 8.6.

CC2 (Compensation absorption): Let $\bar{\sigma}$ be federally invalid and e an event targeting R_m . Source normal forms are determined by component CC2 (if e targets a source) or unaffected (otherwise). Shared components of non-sources are deterministic functions of source normal forms, which agree between the two CC2 paths. Local components of non-sources converge by component CC2 (morphism repair does not modify local components).

CC1 (Order independence): Let e_1, e_2 be independent. If both target the same source, component CC1 gives agreement; if they target different registries, each source's normal form depends only on its own events. Shared components follow from source agreement via Lemma 8.6(b). Local components: if both target the same non-source, component CC1 gives agreement; otherwise each local component sees at most one event. □ □

Theorem 8.9 (Federated Normalization Confluence). *Let $\mathcal{N}$ be a tree-shaped morphism network where each morphism satisfies M1 and each component registry satisfies WFC and CC. Then any two processors consuming the same federated events converge to the same federally valid state.*

Proof. By Lemma 8.7, ρ_{Fed} is a one-shot normalizer: it maps any federated state to a federally valid state in a single application (since morphism repair modifies only shared components and M1 preserves local validity, Phase 2 cannot increase the compensation measure of any component). The induced rewrite system satisfies WFC with trivial federated measure $\Phi_{\text{Fed}}(\bar{\sigma}) = \mathbf{1}[V_{\text{Fed}}(\bar{\sigma}) = \perp] \in \{0, 1\}$; a single compensation step always reaches validity. Termination of the full rewrite system (apply + compensate) follows from the lexicographic order $(|B|, \Phi_{\text{Fed}}(\bar{\sigma}))$. By Theorem 8.8, the system satisfies CC. Theorem 5.5 gives unique normal forms. □ □

Corollary 8.10 (Constructive Normal Form). *The federated normal form is constructive: compute each source’s single-registry normal form independently, propagate shared components through morphisms in topological order, and let each non-source converge its local component via component CC.*

Remark 8.11 (Compensation Compatibility is Derived). One might expect morphisms to satisfy $\phi(\rho_A^*(\sigma)) = \rho_B^*(\phi(\sigma))$. The authority argument shows this is unnecessary: ρ_{Fed} normalizes locally first, then projects via ϕ . The composition “normalize source, then project” produces a valid shared component by M1 alone. We never need the reverse.

Remark 8.12 (Relationship to Product Composition). Theorem 10.13 handles the special case $E = \emptyset$: independent registries with no morphism constraints. Federated convergence generalizes product composition to the case where registries are connected by directed semantic constraints.

8.4 Necessity of the Structural Conditions

Proposition 8.13 (Necessity of Acyclicity). *There exists a cyclic morphism network where each component satisfies WFC and CC, all morphisms satisfy M1, but federated compensation cycles.*

Proof. $R_A = R_B$: $\Sigma = \{0, 1\}$, both valid, trivial compensation. Morphisms: $\phi_{AB}(x) = 1 - x$ (flip), $\phi_{BA} = \text{id}$. Both satisfy M1. From $(0, 0)$: repair on ϕ_{AB} gives $(0, 1)$; repair on ϕ_{BA} gives $(1, 1)$; repair on ϕ_{AB} gives $(1, 0)$; repair on ϕ_{BA} gives $(0, 0)$. Cycle. $\square$ $\square$

Proposition 8.14 (Necessity of M1). *There exists an acyclic network where each component satisfies WFC and CC, but a morphism violates M1, and federated compensation oscillates.*

Proof. R_A : $\Sigma_A = \{0\}$, trivial. R_B : $\Sigma_B = \{0, 1\}$, $V_B(0) = \top$, $V_B(1) = \perp$, $\rho_B(1) = 0$. Total morphism $\phi_{AB}(0) = 1$ (violates M1). From $(0, 0)$: morphism repair sets $\sigma_B = 1$; local compensation sets $\sigma_B = 0$; morphism repair sets $\sigma_B = 1$. Oscillation between morphism repair and local compensation. $\square$ $\square$

Remark 8.15 (The Multi-Source Boundary). The tree-shaped restriction excludes networks where a registry has multiple incoming morphisms from independent sources. When two morphisms ϕ_{ij} and ϕ_{kj} both target registry j , a federally valid state requires $\phi_{ij}(\sigma_i) = \phi_{kj}(\sigma_k) = \pi_j^{\text{sh}}(\sigma_j)$ —constraining the *sources* to produce compatible projections. The authority argument breaks: no single source determines the target. Resolving this requires additional machinery (composition consistency along common ancestors, target agreement predicates, or conflict resolution operators). We develop the last of these—conflict resolution operators—in Section 8.5, recovering convergence on any acyclic network.

8.5 Multi-Source Resolution

Remark 8.15 identified the obstruction to networks in which a registry has several incoming morphisms: no single source determines the target’s shared component. We remove the tree restriction—admitting any acyclic network—by replacing the per-edge authority function at a multi-source target with a single *resolution operator*. The authority argument then generalizes: a target’s shared component becomes a deterministic function not of one source but of all of them.

Definition 8.16 (Resolution Operator). Let j be a registry with source set $S(j) = \{i : (i, j) \in E\}$. A *resolution operator* for j is a map

$$R_j : \prod_{i \in S(j)} \Sigma_{R_i} \longrightarrow S_j$$

from the source states to j 's shared component, satisfying:

- (R1) **Source determinacy:** R_j is a function of the source states alone; it does not read $\pi_j^{\text{loc}}(\sigma_j)$.
- (R2) **Validity preservation:** for all $(\sigma_i)_{i \in S(j)}$ with each $V_{R_i}(\sigma_i) = \top$ and all σ_j with $V_{R_j}(\sigma_j) = \top$,

$$V_{R_j}(\sigma_j[\text{sh} \mapsto R_j((\sigma_i)_{i \in S(j)})]) = \top.$$

When $|S(j)| = 1$, taking $R_j(\sigma_i) = \phi_{ij}(\sigma_i)$ recovers the authority morphism and (R2) is exactly M1. A resolution operator thus strictly generalizes the single-source morphism image: it expresses merges—priority, conjunction, most-restrictive-wins—that no single authority can. As with M1, (R2) is nonvacuous: WFC (Axiom 3.6) on each source and on j makes both antecedents satisfiable.

Definition 8.17 (Resolved Network). A *resolved network* is an acyclic morphism network in which every target with $|S(j)| > 1$ carries a resolution operator R_j ; single-source targets retain their authority morphism. The federated registry is as in Definition 8.4, with the morphism invariants for a multi-source target replaced by the *resolution invariant*

$$\mu_j(\bar{\sigma}) = \top \iff \pi_j^{\text{sh}}(\sigma_j) = R_j((\sigma_i)_{i \in S(j)}).$$

The normalization operator ρ_{Fed} carries over from Definition 8.5 unchanged, with Phase 2 setting each multi-source target's shared component in a single step: for target j in topological order, if $\pi_j^{\text{sh}}(\sigma_j) \neq R_j((\sigma_i)_{i \in S(j)})$, set $\sigma_j \mapsto \sigma_j[\text{sh} \mapsto R_j((\sigma_i)_{i \in S(j)})]$. Acyclicity guarantees a topological ordering in which every source $i \in S(j)$ precedes j .

Lemma 8.18 (Single-Round Termination, Resolved). *On a resolved network, a single application of ρ_{Fed} produces a federally valid state.*

Proof. Identical to Lemma 8.7, with the single-source repair $\phi_{ij}(\sigma_i)$ replaced by $R_j((\sigma_i)_{i \in S(j)})$ and M1 by (R2). Phase 1 establishes local validity; Phase 2 sets each target's shared component exactly once—its resolution operator consumes all incoming edges in one evaluation, and sources are finalized first by acyclicity. By (R2), overwriting a locally valid target's shared component with the resolved value of locally valid sources preserves target validity, so Phase 2 never reintroduces invalidity. $\square$ $\square$

Theorem 8.19 (Federated Convergence with Resolution). *Let $\mathcal{N}$ be an acyclic resolved network in which every single-source morphism satisfies M1, every resolution operator satisfies (R1)–(R2), and every component registry satisfies WFC and CC. Then any two processors consuming the same federated events converge to the same federally valid state.*

Proof. The three-component decomposition of Lemma 8.6 generalizes. (a) Source registries (in-degree zero) are untouched by resolution repair and equal their single-registry normal forms. (b) In a federally valid state, a target j satisfies $\pi_j^{\text{sh}}(\sigma_j) = R_j((\sigma_i)_{i \in S(j)})$ (or $\phi_{ij}(\sigma_i)$ when $|S(j)| = 1$)—a deterministic function of its sources' normal forms. (c) Its local component is untouched by resolution (by (R1) the operator writes only the shared component) and converges by component CC.

By induction on the topological order, each component's normal form is a deterministic function of the per-component event multisets. *CC2*: for a federally invalid $\bar{\sigma}$ and event e , source normal forms are fixed by component CC2 or unaffected; each shared component is a deterministic function (via R_j or ϕ) of source normal forms, which agree across the two paths; local components converge by component CC2. *CC1*: independent events targeting distinct registries affect disjoint per-component multisets, and each shared component is recomputed from source normal forms, so order is immaterial; independent events on a common registry commute by component CC1. Hence the federated system satisfies CC. Lemma 8.18 gives WFC with the trivial measure $\Phi_{\text{Fed}} \in \{0, 1\}$, so Theorem 5.5 yields a unique normal form. $\square$ $\square$

Corollary 8.20 (Constructive Resolved Normal Form). *Corollary 8.10 extends: compute each source's single-registry normal form, then, in topological order, set each target's shared component by evaluating its resolution operator (or authority morphism) on its finalized sources, and let each target converge its local component by component CC. As in the tree case, the product state space is never materialized.*

Remark 8.21 (Verified, not assumed). Unlike the authority morphism, a resolution operator encodes domain policy: there is no canonical merge. The theorem is therefore conditioned on (R1)–(R2), exactly as the single-source case is conditioned on M1. Both conditions are decidable—the source and target state spaces are finite, so (R1) and (R2) are checked by enumerating $\prod_{i \in S(j)} \Sigma_{R_i} \times \Sigma_{R_j}$ per multi-source target—so a build-time verifier certifies a resolved network exactly as it certifies WFC and CC for a single registry. Theorem 8.9 is the special case in which every target has a single source whose resolution operator is the morphism image.

Remark 8.22 (Necessity of the resolution conditions). Acyclicity remains necessary (Proposition 8.13 is unchanged; resolution operators do not tame cycles). (R2) is necessary for the reason M1 is (Proposition 8.14): a resolver mapping a valid source combination to a shared component that violates V_{R_j} induces oscillation between resolution repair and local compensation. (R1) is necessary for well-definedness: were R_j to read the local component, the shared normal form would depend on the order in which local compensation and resolution repair interleave, defeating single-round termination.

8.6 Monotone Cycles

Acyclicity (Definitions 8.3 and 8.17) is required because the necessity counterexamples turn on *non-monotone* repair: the cyclic counterexample (Proposition 8.13) uses the negation morphism $\phi(x) = 1 - x$, and the M1 counterexample (Proposition 8.14) an antitone flip. We show that when repair is *monotone* on an ordered shared domain, acyclicity is unnecessary: convergence follows from the Knaster–Tarski fixed-point theorem, and order-independence is exactly the classical convergence of *chaotic iteration*.

Definition 8.23 (Ordered Shared Domains). A network has *ordered shared domains* if each shared space S_j carries a partial order $\sqsubseteq_j$ making $(S_j, \sqsubseteq_j)$ a complete lattice (least element $\perp_j$; all joins exist). With the component local normal forms fixed (order-independent by component CC), define the *federated repair operator* $\Phi : \prod_j S_j \rightarrow \prod_j S_j$ by

$$\Phi(\bar{s})_j = \begin{cases} s_j^{\text{ev}} & j \text{ a source (no incoming): its event-determined shared value,} \\ \phi_{ij}(\sigma_i[\text{sh} \mapsto s_i]) & j \text{ single-source from } i, \\ R_j((\sigma_i[\text{sh} \mapsto s_i])_{i \in S(j)}) & j \text{ multi-source.} \end{cases}$$

The morphisms and resolvers are *monotone* when Φ is monotone: $\bar{s} \sqsubseteq \bar{s}' \Rightarrow \Phi(\bar{s}) \sqsubseteq \Phi(\bar{s}')$ (componentwise). Source entries are constant maps, hence monotone; the condition constrains only the morphism images and resolvers.

Theorem 8.24 (Monotone Convergence Despite Cycles). *Let $\mathcal{N}$ be a morphism network with ordered shared domains (Definition 8.23), possibly cyclic, whose morphisms and resolvers are monotone and validity-preserving (M1 / R2), and whose components satisfy WFC and CC. Then Φ has a least fixed point $\bar{s}^* = \bigsqcup_{k \geq 0} \Phi^k(\perp)$, and every fair asynchronous (any-order) iteration of the component updates from $\perp$ converges to $\bar{s}^*$. Consequently all processors consuming the same federated events converge to the same federally valid state, with no acyclicity requirement.*

Proof. Φ is a monotone self-map of the complete lattice $\prod_j S_j$ (a product of complete lattices), so by Knaster–Tarski [14] it has a least fixed point, obtained by Kleene iteration as $\bar{s}^* = \bigsqcup_k \Phi^k(\perp)$ —the ascending chain $\perp \sqsubseteq \Phi(\perp) \sqsubseteq \dots$ stabilizes (for a finite lattice in at most its height times $|\mathcal{I}|$ steps). Order-independence is the convergence of chaotic iteration: for a monotone operator on a complete lattice, any fair schedule of component updates started from $\perp$ reaches the same least fixed point [15]. Validity is preserved along the iteration by M1/R2 exactly as in Lemma 8.18, so $\bar{s}^*$ is federally valid. With the local components fixed by component CC, the full federated normal form is determined; hence the system is confluent. $\square \quad \square$

Corollary 8.25 (Acyclicity is a concession to non-monotonicity). *Acyclicity is needed only to tame non-monotone repair. The cyclic divergence of Proposition 8.13 uses the antitone negation $\phi(x) = 1 - x$; under monotonicity that operator is excluded and Theorem 8.24 applies. Acyclic structure and monotonicity are thus two independent routes to federated confluence—the former for arbitrary (possibly non-monotone) repair, the latter for arbitrary (possibly cyclic) topology.*

Remark 8.26 (Order-independence is chaotic iteration). Theorem 8.24 exposes the central property of normalization confluence in another guise: convergence “regardless of application order” is precisely the order-independent convergence of chaotic iteration in dataflow analysis and abstract interpretation [15]. The monotone regime inherits that machinery—worklist scheduling, and widening/narrowing for tall or infinite lattices—for computing and accelerating the federated normal form.

Remark 8.27 (CRDTs are compensation-free monotone federations). A join-semilattice whose merge is its join is monotone and trivially validity-preserving, so a state-based CRDT [3] is the special case of Theorem 8.24 in which repair never has to restore an invariant. Normalization confluence adds compensation on top: monotone convergence *with* business invariants, which operation commutativity alone does not provide.

Remark 8.28 (Infinite lattices and widening). Theorem 8.24 assumes only a *complete* lattice, which may be infinite. Knaster–Tarski still yields the least fixed point $\bar{s}^*$; what changes is its computation. Kleene iteration reaches $\bar{s}^*$ in finitely many steps under the ascending chain condition (in particular, for any finite lattice). Without ACC—an unbounded counter or resource in $\mathbb{N} \cup \{\top\}$, whose ascending chains never stabilize—iteration need not terminate; a *widening* operator (abstract interpretation [15]) forces termination by extrapolating the ascending chain to a sound post-fixed-point, trading exactness for computability. The convergence guarantee is unaffected: the federated normal form exists and is unique regardless; only its exact computation requires ACC or a widening.

8.7 Compositionality: Sub-Federations as Effective Registries

Product composition (Theorem 10.13) collapses *independent* registries into one. The same holds for a *connected* sub-federation: an internally convergent sub-network can be replaced by a single *effective registry*, so federations verify modularly and hierarchically. Registries and morphisms form a category in which the federated registry is the limit of the constraint diagram; this subsection is the statement that such limits compose.

Definition 8.29 (Sub-Federation, Boundary, Convexity). Given $\mathcal{N} = (\{R_i\}_{i \in \mathcal{I}}, \{\phi_{ij}\})$ and $\mathcal{J} \subseteq \mathcal{I}$, the *sub-federation* on $\mathcal{J}$ has components $\{R_i\}_{i \in \mathcal{J}}$ and *internal* morphisms $E_{\mathcal{J}} = \{(i, j) \in E : i, j \in \mathcal{J}\}$. Its boundary consists of *import* edges (from $\mathcal{I} \setminus \mathcal{J}$ into $\mathcal{J}$) and *export* edges (from $\mathcal{J}$ out). $\mathcal{J}$ is *convex* if no directed path between two members of $\mathcal{J}$ passes through a vertex outside $\mathcal{J}$.

Definition 8.30 (Effective Registry). The *effective registry* $R_{\mathcal{J}}$ is the federated registry (Definition 8.4) of the sub-federation: state space $\prod_{i \in \mathcal{J}} \Sigma_{R_i}$, invariants the internal local and morphism invariants, and compensation the sub-federated normalizer $\rho_{\text{Fed}}^{\mathcal{J}}$. Its shared component is the union of the import ports (the shared components of members targeted by import edges); export edges read its normalized state.

Theorem 8.31 (Compositional Collapse). *Let $\mathcal{J}$ be convex with an internally convergent sub-federation (its morphisms satisfy M1/R2 and its components satisfy WFC and CC; acyclic, or cyclic with monotone repair). Then:*

- (a) $R_{\mathcal{J}}$ satisfies WFC and CC.
- (b) Each boundary morphism, reinterpreted as a morphism to or from $R_{\mathcal{J}}$, satisfies M1 iff it does in $\mathcal{N}$.
- (c) Let $\mathcal{N}'$ replace the sub-federation by $R_{\mathcal{J}}$. Then $\mathcal{N}'$ is a well-formed network, acyclic if $\mathcal{N}$ is, and $\mathcal{N}$ converges iff $\mathcal{N}'$ does; their normal forms agree under $\Sigma_{R_{\mathcal{J}}} = \prod_{i \in \mathcal{J}} \Sigma_{R_i}$.

Proof. (a) WFC: $\rho_{\text{Fed}}^{\mathcal{J}}$ drives the sub-federated measure to a federally valid state (Lemma 8.7, Lemma 8.18), so it is a well-founded compensation for $R_{\mathcal{J}}$. CC: internal convergence (Theorem 8.9, Theorem 8.19, or Theorem 8.24) makes the sub-federated normal form a deterministic function of the events targeting $\mathcal{J}$ and the import-port values; CC1 (order independence of internal events) and CC2 (absorption of internal compensation) are precisely that confluence read through $R_{\mathcal{J}}$.

(b) An import edge (i, j) with $j \in \mathcal{J}$ overwrites the shared component of $R_{\mathcal{J}}$ that coincides with π_j^{sh} ; its M1 obligation—overwriting with a valid source image preserves target validity—is unchanged, since sub-federated re-normalization preserves federal validity by (a). An export edge reads $R_{\mathcal{J}}$ ’s normalized state, a source-determined projection.

(c) Convexity ensures contracting $\mathcal{J}$ to a single vertex creates no exterior path between two members, so $\mathcal{N}'$ is acyclic whenever $\mathcal{N}$ is. With (a)–(b), $\mathcal{N}'$ meets the hypotheses of the federated convergence theorem. Because $\mathcal{J}$ is convex, a topological order of $\mathcal{N}$ visits it as a contiguous block, so ρ_{Fed} on $\mathcal{N}$ factors as “normalize the sub-federation, then propagate across the boundary”—which is exactly ρ_{Fed} on $\mathcal{N}'$ with $R_{\mathcal{J}}$ ’s compensation inlined. Hence the normal forms coincide. $\square$ $\square$

Corollary 8.32 (Modular and Hierarchical Verification). *A federation verifies compositionally: verify each convex sub-federation once, replace it by its effective registry, and verify the network of effective registries. The construction nests—giving hierarchical verification—and a verified sub-federation is reusable as a black-box component with a fixed port interface. Theorem 10.13 is the special case $E_{\mathcal{J}} = \emptyset$ with empty boundary (independent registries).*

Remark 8.33 (Convexity and the monotone regime). Convexity is needed only to keep the collapsed network acyclic. In the monotone regime (Theorem 8.24) a boundary-induced cycle is harmless: if the sub-federation’s repair is monotone then so is $\rho_{\text{Fed}}^{\mathcal{J}}$, and Theorem 8.24 applies to $\mathcal{N}'$ regardless of convexity.

8.8 Example: Manufacturer–Supplier Federation

Manufacturer R_M : $\Sigma_M = \{\text{draft}, \text{active}, \text{suspended}\}$, all locally valid. $e_{\text{pub}}: \text{draft} \rightarrow \text{active}$.

Supplier R_S : $\Sigma_S = \{\text{idle}, \text{listed}, \text{stale}, \text{err}\}$; $V_S(\text{err}) = \perp$, rest valid; $\rho_S(\text{err}) = \text{idle}$. e_{exp} : sets state to **stale**.

Total morphism ϕ : $\text{draft} \mapsto \text{idle}$, $\text{active} \mapsto \text{listed}$, $\text{suspended} \mapsto \text{stale}$. M1 holds (all images are valid).

Convergence trace from $(\text{draft}, \text{idle})$ with independent events e_{pub} (targets R_M) and e_{exp} (targets R_S):

Order 1 (e_{pub} first): $(\text{active}, \text{idle}) \xrightarrow{\phi} (\text{active}, \text{listed}) \xrightarrow{e_{\text{exp}}} (\text{active}, \text{stale}) \xrightarrow{\phi} (\text{active}, \text{listed})$.

Order 2 (e_{exp} first): $(\text{draft}, \text{stale}) \xrightarrow{\phi} (\text{draft}, \text{idle}) \xrightarrow{e_{\text{pub}}} (\text{active}, \text{idle}) \xrightarrow{\phi} (\text{active}, \text{listed})$.

Both converge to $(\text{active}, \text{listed})$. The supplier event’s effect is overwritten by morphism repair: the manufacturer is authoritative.

9 Discussion

9.1 Three Regimes of Coordination-Free Convergence

Our result, combined with CRDTs [3] and invariant confluence [6], identifies three distinct structural regimes under which distributed processors converge without coordination:

1. **Operation commutativity** (CRDTs): Operations commute algebraically. Convergence follows from the algebra of operations. No invariant enforcement or compensation is needed.
2. **Invariant confluence** (I-confluence): Operations preserve application invariants under all orderings. Convergence follows because validity is never violated. No compensation is needed.
3. **Normalization confluence** [1]: Operations may be non-commutative and may violate invariants. Compensation normalizes states to validity. WFC and CC are sufficient for semantic convergence (Theorem 5.5); UBC is orthogonal to convergence and governs scalability (Theorem 7.1). No coordination is needed.

The three regimes form a containment hierarchy. Operation commutativity implies CC (with $\rho_R^* = \text{id}$). Invariant confluence implies that compensation never fires. Normalization confluence subsumes both as special cases and additionally covers systems where operations are non-commutative and invariant-violating—the common case in business applications.

The regimes partition the coordination-free design space by *where* convergence is guaranteed: in the operations (regime 1), in the invariants (regime 2), or in the normalization rewrite system (regime 3).

Remark 9.1 (Convergence as Rewrite Property). The conventional literature frames convergence as a property of operations: operations must commute, or operations must preserve invariants. Normalization confluence reframes convergence as a property of the *rewrite system* that interleaves application and compensation. The operations themselves need no special algebraic structure. This shifts the design burden from operation algebra to compensation design, and places the convergence mechanism at the semantic normalization layer—above the data-structure layer (CRDTs) and the transaction layer (I-confluence). Moreover, this rewrite-system

perspective extends naturally across organizational boundaries: the federated convergence result (Section 8) shows that multi-registry systems inherit convergence from their components, with morphism repair as additional rewrite rules in the same framework.

9.2 Two Regimes for Federated Repair

The federated results partition constraint-network design into two regimes with *independent* routes to convergence (Corollary 8.25). When repair is *monotone* on ordered shared domains, convergence holds on *any* topology—including cyclic meshes—by Theorem 8.24. When repair may be *non-monotone* (toggles, cancellations, order-reversing revocations), the topology must be *acyclic*, with resolution operators (Section 8.5) at multi-source targets, and convergence is the one-shot constructive normal form (Corollary 8.20). In both regimes repair must preserve validity (M1/R2). Table 1 summarizes the trade; the two axes—monotonicity and acyclicity—are orthogonal, and either alone suffices.

Regime	Repair requirement	Topologies	Convergence guarantee
Monotone	Monotone on a lattice order (join or meet); validity-preserving	Any: cyclic meshes, DAGs, trees	Least fixed point $\text{lfp}(\perp)$ under any <i>fair</i> asynchronous order (Knaster–Tarski + chaotic iteration)
Non-monotone	May reverse the order (toggles, negation, cancellations); validity-preserving	Acyclic only; multi-source targets need resolution operators	Deterministic one-shot constructive normal form; feedback loops structurally excluded

Table 1: Two regimes for federated repair. Monotonicity and acyclicity are independent conditions for confluence: monotone repair converges on any topology (Theorem 8.24); non-monotone repair requires acyclicity (Proposition 8.13). Validity preservation (M1/R2) is required in both.

9.3 Verifying Compensation Commutativity

CC is easy to state but potentially hard to verify: it quantifies over all states and all independent event pairs. Section 10 develops a verification calculus that reduces CC to local, per-event-pair checks via invariant footprints and decomposable repair. Here we note three common patterns that practitioners can use as a first check before invoking the full calculus:

1. **Compensation to a canonical state.** If there exists a default valid state $\sigma_\perp$ such that $\rho_R^*(\sigma) = \sigma_\perp$ for all invalid σ , both CC1 and CC2 hold trivially. This is the “reset to safe default” pattern.
2. **Lattice-structured compensation.** If the valid states form a join-semilattice and compensation computes the join with the nearest valid state, both conditions hold by the commutativity of the join operation.
3. **Decomposable repair.** If compensation decomposes into commuting per-invariant repairs, CC can be verified via the footprint calculus of Section 10. The order fulfillment example (Section 4) uses this pattern.

9.4 Applications

The compensate-and-converge principle applies wherever systems process streams of changes under semantic constraints with bounded compensation. We highlight three domains.

Infrastructure Configuration Drift. Infrastructure-as-code tools apply streams of configuration changes to infrastructure state. Valid configurations are defined by policy. The semantic state space is finite (k boolean policy predicates yield at most 2^k equivalence classes). Remediation operators that reset violated policies to compliant defaults satisfy CC2 (compensation to a canonical state per policy). Our theorem guarantees convergence to equivalent policy-compliant states regardless of the order in which changes are applied.

Database Schema Migration. Schema migrations are event streams applied to schema state. The set of valid schema configurations is finite. Migration scripts that repair constraint violations constitute compensation operators. When such scripts resolve each constraint independently (lattice-structured compensation), CC holds and our theorem guarantees order-independent convergence.

Regulatory Compliance. Organizations processing streams of regulatory changes must maintain compliance. Compliance states are finite (each regulation is either satisfied or not). Remediation procedures that restore compliance by addressing each violated regulation independently satisfy CC. Our theorem guarantees convergence to equivalent compliance states.

Cross-Organizational Supply Chains. The federated convergence result (Section 8) applies to multi-registry settings. A manufacturer’s product specification registry connected by morphisms to supplier capability registries forms a tree-shaped network. Each organization maintains its own compensation operators; the morphisms encode specification conformance. Theorem 8.9 guarantees that all participants converge to consistent states across organizational boundaries. When a supplier answers to *several* authorities at once—say a manufacturer’s specifications *and* a regulator’s rules—the network is no longer a tree; a resolution operator (Section 8.5) combines the constraints (for instance, most-restrictive-wins), and Theorem 8.19 extends the same guarantee to this acyclic multi-source setting.

9.5 Extensions

One extension remains future work: *privacy-preserving validation* where organizations prove registry compliance via zero-knowledge proofs without revealing state. The federated convergence result (Section 8) handles the multi-registry coordination problem; privacy-preserving validation would additionally protect organizational sovereignty over state visibility.

10 Verification Calculus for CC

Compensation commutativity is a global condition: it quantifies over all states and all independent event pairs. This section, based on [1], develops sufficient conditions that reduce CC to local, per-event-pair checks. The key tools are an *invariant footprint* mapping and a *decomposable repair* condition.

10.1 Notation

For readability in algebraic manipulations, we write $N(\sigma) := \rho_R^*(\sigma)$ for iterated compensation to validity (Definition 3.15). The normalizer satisfies: (i) $V_R(N(\sigma)) = \top$ for all σ ; (ii) $N(\sigma) = \sigma$ if $V_R(\sigma) = \top$; (iii) $N(N(\sigma)) = N(\sigma)$ (idempotence). We write $A_e(\sigma) := \text{apply}(e, \sigma)$ when convenient.

In this notation, the CC obligations become:

- (CC1) $N(A_{e_2}(N(A_{e_1}(\sigma)))) = N(A_{e_1}(N(A_{e_2}(\sigma))))$ for independent e_1, e_2 .
- (CC2) $N(A_e(\sigma)) = N(A_e(\rho_R(\sigma)))$ for invalid σ .

10.2 Invariant Vectors and Footprints

Definition 10.1 (Invariant Vector). Define $\mathbf{v} : \Sigma_R \rightarrow \{0, 1\}^k$ by $\mathbf{v}(\sigma)_i = 1 \iff \psi_i(\sigma) = \top$.

Definition 10.2 (Footprint). A *footprint* is a function $F : \mathcal{E} \rightarrow 2^{\{1, \dots, k\}}$ such that for all σ and all $i \notin F(e)$,

$$\mathbf{v}(A_e(\sigma))_i = \mathbf{v}(\sigma)_i.$$

Intuitively, $F(e)$ is the set of invariants whose truth value can change under e . Over-approximations are sound: larger footprints make the calculus more conservative but never incorrect.

10.3 Decomposable Repair

We capture the common case where repair addresses each violated invariant independently.

Definition 10.3 (Per-Invariant Repair). A set of *per-invariant repair operators* $r_1, \dots, r_k : \Sigma_R \rightarrow \Sigma_R$ satisfies:

- (R1) **Targeted fix:** if $\psi_i(\sigma) = \perp$, then $\psi_i(r_i(\sigma)) = \top$.
- (R2) **No-op on satisfied:** if $\psi_i(\sigma) = \top$, then $r_i(\sigma) = \sigma$.
- (R3) **Mutual commutativity:** $r_i(r_j(\sigma)) = r_j(r_i(\sigma))$ for all i, j .

Definition 10.4 (Decomposable Normalizer). The normalizer N is *decomposable* if there exist per-invariant repairs $r_1, \dots, r_k$ as above such that

$$N(\sigma) = r_1(r_2(\dots r_k(\sigma) \dots))$$

for all σ . (The order is irrelevant by mutual commutativity.)

Definition 10.5 (Repair Locality). An event e is *repair-local* with respect to footprint $F(e)$ if for all $i \notin F(e)$,

$$r_i(A_e(\sigma)) = A_e(r_i(\sigma)).$$

That is, e commutes with repairs of invariants it does not affect.

10.4 Strong Absorption Implies CC2

Theorem 10.6 (Strong Absorption). Assume WFC and let $N = \rho_R^*$. If for all events e and states σ ,

$$N(A_e(\sigma)) = N(A_e(N(\sigma))),$$

then CC2 holds.

Proof. Let $V_R(\sigma) = \perp$. Because N iterates ρ_R to the unique valid fixpoint reachable from σ , and $\rho_R(\sigma)$ lies on the same compensation chain, we have $N(\sigma) = N(\rho_R(\sigma))$.

Applying the assumed equation at σ and at $\rho_R(\sigma)$:

$$N(A_e(\sigma)) = N(A_e(N(\sigma))) = N(A_e(N(\rho_R(\sigma)))) = N(A_e(\rho_R(\sigma))),$$

where the first and third equalities use the assumption and the middle equality uses $N(\sigma) = N(\rho_R(\sigma))$. This is CC2. $\square$

Remark 10.7. Strong absorption states: “normalizing before applying an event does not change the final normalized outcome.” This holds whenever compensation repairs violations in a way determined by the violation pattern rather than by the path to the violation. The order fulfillment example (Section 4) satisfies this: the held state captures the approval intent regardless of whether the order was directly approved with low balance or arrived at the invalid state through a different path.

10.5 Footprint Disjointness Implies CC1

We prove two preparatory lemmas before the main theorem.

Lemma 10.8 (Repair-Event Commutation). *If event e is repair-local with respect to footprint $F(e)$ (Definition 10.5), then for any index set J with $J \cap F(e) = \emptyset$ and any state σ :*

$$R_J(A_e(\sigma)) = A_e(R_J(\sigma)),$$

where $R_J = \prod_{i \in J} r_i$.

Proof. By repair locality, $r_i \circ A_e = A_e \circ r_i$ for each $i \in J$ (since $i \notin F(e)$). The result follows by induction on $|J|$: if the claim holds for $R_{J'}$ with $|J'| < |J|$, then for any $j \in J$,

$$R_J(A_e(\sigma)) = r_j(R_{J \setminus \{j\}}(A_e(\sigma))) = r_j(A_e(R_{J \setminus \{j\}}(\sigma))) = A_e(r_j(R_{J \setminus \{j\}}(\sigma))) = A_e(R_J(\sigma)),$$

using the inductive hypothesis and then $r_j \circ A_e = A_e \circ r_j$. $\square$

Lemma 10.9 (Repair Block Idempotence). *For any index set $J \subseteq \{1, \dots, k\}$, the repair block R_J is idempotent on states that are valid on all invariants in J : if $\psi_i(\sigma) = \top$ for all $i \in J$, then $R_J(\sigma) = \sigma$. More generally, $R_J(R_J(\sigma)) = R_J(\sigma)$ for all σ .*

Proof. If $\psi_i(\sigma) = \top$ for all $i \in J$, then each $r_i(\sigma) = \sigma$ by (R2), so $R_J(\sigma) = \sigma$. For general idempotence: $R_J(\sigma)$ satisfies $\psi_i(R_J(\sigma)) = \top$ for all $i \in J$ (each r_i fixes its invariant by (R1) and does not break others by mutual commutativity and (R2)), so $R_J(R_J(\sigma)) = R_J(\sigma)$ by the first claim. $\square$

Theorem 10.10 (Footprint Commutativity). *Assume WFC and that N is decomposable via commuting per-invariant repairs $r_1, \dots, r_k$. Let e_1, e_2 be independent events satisfying:*

- (H1) *Disjoint footprints: $F(e_1) \cap F(e_2) = \emptyset$.*
- (H2) *Repair locality: each e_j is repair-local with respect to $F(e_j)$.*

Then CC1 holds for (e_1, e_2) .

Proof. Partition $\{1, \dots, k\}$ into three disjoint index sets: $S_1 = F(e_1)$, $S_2 = F(e_2)$, $S_0 = \{1, \dots, k\} \setminus (S_1 \cup S_2)$. By decomposability and mutual commutativity, $N = R_{S_0} \circ R_{S_1} \circ R_{S_2}$ in any order. We compute both sides of CC1 as chains of equalities.

Left side: $N(A_{e_2}(N(A_{e_1}(\sigma))))$.

$$\begin{aligned}
&= N(A_{e_2}(R_{S_0}(R_{S_1}(R_{S_2}(A_{e_1}(\sigma))))) && \text{(expand inner } N) \\
&= N(R_{S_0}(R_{S_1}(A_{e_2}(R_{S_2}(A_{e_1}(\sigma))))) && \text{(Lemma 10.8: } S_0 \cap F(e_2) = S_1 \cap F(e_2) = \emptyset) \\
&= R_{S_0}(R_{S_1}(R_{S_2}(R_{S_0}(R_{S_1}(A_{e_2}(R_{S_2}(A_{e_1}(\sigma))))) && \text{(expand outer } N) \\
&= R_{S_0}(R_{S_1}(R_{S_2}(A_{e_2}(R_{S_2}(A_{e_1}(\sigma))))) && \text{(Lemma 10.9: } R_{S_0}, R_{S_1} \text{ absorbed)}
\end{aligned}$$

Right side: $N(A_{e_1}(N(A_{e_2}(\sigma))))$.

$$\begin{aligned}
&= N(A_{e_1}(R_{S_0}(R_{S_2}(R_{S_1}(A_{e_2}(\sigma))))) && \text{(expand inner } N) \\
&= N(R_{S_0}(R_{S_2}(A_{e_1}(R_{S_1}(A_{e_2}(\sigma))))) && \text{(Lemma 10.8: } S_0 \cap F(e_1) = S_2 \cap F(e_1) = \emptyset) \\
&= R_{S_0}(R_{S_1}(R_{S_2}(R_{S_0}(R_{S_2}(A_{e_1}(R_{S_1}(A_{e_2}(\sigma))))) && \text{(expand outer } N) \\
&= R_{S_0}(R_{S_1}(R_{S_2}(A_{e_1}(R_{S_1}(A_{e_2}(\sigma))))) && \text{(Lemma 10.9: } R_{S_0}, R_{S_2} \text{ absorbed)}
\end{aligned}$$

Equating. The left side is $R_{S_0}(R_{S_1}(R_{S_2}(A_{e_2}(R_{S_2}(A_{e_1}(\sigma)))))$ and the right side is $R_{S_0}(R_{S_1}(R_{S_2}(A_{e_1}(R_{S_1}(A_{e_2}(\sigma)))))$. It remains to show their inner expressions are equal.

Apply Lemma 10.8 to move R_{S_2} past A_{e_1} on the left (since $S_2 \cap F(e_1) = \emptyset$):

$$A_{e_2}(R_{S_2}(A_{e_1}(\sigma))) = A_{e_2}(A_{e_1}(R_{S_2}(\sigma))).$$

Apply Lemma 10.8 to move R_{S_1} past A_{e_2} on the right (since $S_1 \cap F(e_2) = \emptyset$):

$$A_{e_1}(R_{S_1}(A_{e_2}(\sigma))) = A_{e_1}(A_{e_2}(R_{S_1}(\sigma))).$$

Both inner expressions now have the form $A_{e_j}(A_{e_\ell}(R_{S_?}(\sigma)))$ with all repairs pushed past the events. Applying R_{S_2} to the left expression and R_{S_1} to the right, then using mutual commutativity of R_{S_1} and R_{S_2} , both sides reduce to

$$R_{S_0}(R_{S_1}(R_{S_2}(A_{e_2}(A_{e_1}(R_{S_2}(R_{S_1}(\sigma)))))$$

Since R_{S_1} and R_{S_2} commute and each is idempotent after application (Lemma 10.9), the left and right sides are equal. $\square$

Remark 10.11 (Practical import). A practitioner does not prove CC1 by global state enumeration. Instead: (i) assign each event type an invariant footprint, (ii) verify that each event commutes with repairs outside its footprint (Lemma 10.8), (iii) check that independent event pairs have disjoint footprints. All three are local obligations. The order fulfillment example (Section 4) demonstrates this workflow: e_{credit} and e_{approve} affect disjoint aspects of the state, and the held-state repair commutes with credit application.

10.6 Product Composition

Real systems comprise multiple semantic modules. The following theorem shows CC verifications compose.

Definition 10.12 (Product Registry). Let $R_1 = (\Sigma_1, I_1, V_1, \rho_1)$ and $R_2 = (\Sigma_2, I_2, V_2, \rho_2)$ with disjoint invariant sets. Define $R_1 \otimes R_2$ by:

$$\Sigma = \Sigma_1 \times \Sigma_2, \quad V(\sigma_1, \sigma_2) = V_1(\sigma_1) \wedge V_2(\sigma_2), \quad \rho(\sigma_1, \sigma_2) = (\rho_1(\sigma_1), \rho_2(\sigma_2)).$$

Events are pairs $e = (e^{(1)}, e^{(2)})$ with application componentwise.

Theorem 10.13 (Product Lifting). *If each R_j satisfies WFC and CC, then $R_1 \otimes R_2$ satisfies WFC and CC. If each normalizer is decomposable, the product normalizer is decomposable with footprints composed by union across components.*

Proof. WFC: Use measure $\Phi(\sigma_1, \sigma_2) = \Phi_1(\sigma_1) + \Phi_2(\sigma_2)$. Each compensation step strictly decreases at least one component when invalid.

CC2: The product normalizer satisfies $N(\sigma_1, \sigma_2) = (N_1(\sigma_1), N_2(\sigma_2))$ and application is componentwise, so CC2 holds iff it holds in each component.

CC1: Both sides of the CC1 equation decompose into paired equalities in R_1 and R_2 by componentwise application and normalization.

Decomposability: Per-invariant repairs in each component act on disjoint state components and thus commute across components trivially. $\square$

Remark 10.14. Product composition enables modular system design: verify CC for each semantic module independently, then compose without re-verification. This handles the *independent* case (disjoint invariants, no cross-module constraints). The federated convergence result (Section 8) generalizes product composition to the *dependent* case: registries connected by morphisms encoding cross-organizational semantic constraints.

10.7 Partial Decomposability

Full decomposability is a strong sufficient condition. When it does not hold globally, the calculus still provides value:

1. CC may hold for event pairs with disjoint footprints even if the full normalizer is not decomposable. The footprint theorem (Theorem 10.10) applies to any event pair satisfying its hypotheses, regardless of whether *all* pairs do.
2. When footprints overlap for some event pair, the calculus identifies precisely which pairs require further analysis. A practitioner can then either redesign compensation to be canonical for the overlapping invariants or add coordination only for the interacting event types.
3. For event types that share footprints but whose normalized outcomes agree by domain-specific reasoning (e.g., both repairs clamp to the same bound), CC1 can be verified by case analysis restricted to the overlapping states. The footprint calculus reduces the case space.

11 Conclusion

We have identified *normalization confluence* as a third structural regime for coordination-free convergence in distributed systems, alongside operation commutativity (CRDTs) and invariant confluence (I-confluence). The regime is characterized by two conditions: well-founded compensation (WFC), which ensures termination, and compensation commutativity (CC), which ensures confluence. Under both, the governance rewrite system has unique normal forms by Newman's Lemma, and all processors consuming the same events converge to the same valid

state. WFC and CC are sufficient for convergence; a uniform bound on compensation depth (UBC) is orthogonal to convergence and governs uniform complexity bounds. All three conditions are necessary for their respective guarantees: without UBC, compensation depth is unbounded; without CC, different orders reach different valid states.

The result extends across organizational boundaries. For tree-shaped networks of registries connected by morphisms, federated convergence requires only validity preservation of the morphisms: federated termination and commutativity are derived from the network’s acyclicity and the components’ existing convergence properties, not assumed as additional axioms. The core mechanism is the authority argument: in a directed morphism, the source’s unique normal form deterministically determines the target’s shared component, eliminating the nondeterminism that would otherwise require additional conditions. When a target answers to several sources at once, a *resolution operator* generalizes the authority function—a source-determined, validity-preserving merge—and convergence extends to any acyclic network, with single-source authority as the special case. Because the merge encodes domain policy, its two conditions are stated as decidable hypotheses (checkable by finite enumeration) rather than assumed, mirroring how the single-source case rests on morphism validity preservation. Finally, when the shared domains are ordered lattices and repair is monotone, even *cyclic* networks converge—by the Knaster–Tarski fixed-point theorem—which reveals acyclicity as a requirement only for non-monotone repair, exhibits order-independent convergence as chaotic iteration, and places state-based CRDTs as the compensation-free special case. Federated convergence is moreover *compositional*: a convex, internally convergent sub-federation collapses to a single effective registry, so large networks verify modularly and hierarchically and verified subsystems compose as black-box components.

To make CC practically verifiable, we developed a verification calculus that reduces the global CC obligation to local checks. Invariant footprints identify which invariants each event can affect; decomposable repair ensures per-invariant fixes commute; and product composition enables modular verification. Federated convergence generalizes product composition from independent registries to registries connected by directed semantic constraints.

Compensation commutativity is strictly weaker than the operation commutativity required by CRDTs: operations need not commute, only their compensated results. The gap between these two conditions is precisely the design space that normalization confluence occupies—systems where operations are non-commutative, invariant-violating, but convergent under repair.

The three regimes partition the coordination-free design space by where convergence is guaranteed: in the operations, in the invariants, or in the normalization rewrite system. This last regime formalizes a pattern that practitioners use implicitly—post-application repair as a convergence mechanism—and provides the structural conditions under which it is sound, a calculus for verifying those conditions, and a federated extension for applying them across organizational boundaries.

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